

JPEG AI as a Threat to Deepfake Detection

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Problem Statement

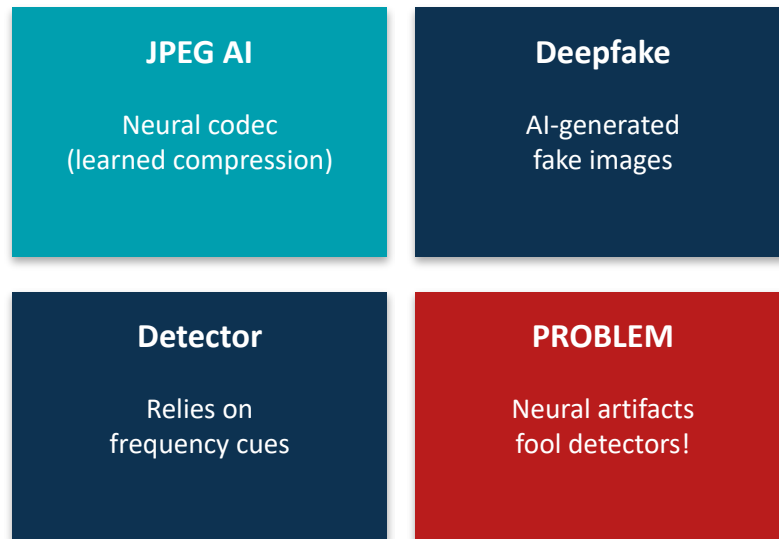
JPEG AI: A New Compression Paradigm

- First international learned image compression standard (ISO/IEC)
- Deep neural networks used for encoding and decoding
- Artifacts with fundamentally different statistical signature than classical JPEG
- Increasingly adopted — threatens existing forensic pipelines

The Forensic Challenge

Deepfake detectors exploit low-level frequency cues and compression artifacts from the generation process.

JPEG AI disrupts these forensic fingerprints, dramatically degrading detector performance.



↓ JPEG AI recompression ↓

Research question:
*How much does JPEG AI compression degrade
deepfake detection performance?*

State of the Art

Deepfake Detection

CLIP-Det (Cozzolino et al., CVPRW 2024): CLIP features + linear classifier.
Corvi2023: ResNet50 fine-tuned on diffusion-generated images.
Detectors exploit low-level statistical fingerprints in the image spectrum.

Classical Compression (JPEG)

Traditional JPEG introduces well-studied block-DCT artifacts.
Detectors degrade gracefully; retraining on JPEG-augmented data restores robustness.
JPEG robustness is a standard forensic benchmark.

JPEG AI & Forensics

Cannas et al. (ICCVW 2025): first study on JPEG AI impact on image forensics.
Neural codecs produce periodicity artifacts from upsampling layers.
These differ fundamentally from block-DCT artifacts => detectors fail unexpectedly.

Mitigation Approaches in Forensics

Preprocessing (filtering) to suppress codec-specific artifacts.
Domain adaptation / fine-tuning on compression-augmented data.
Score fusion to combine complementary detector outputs.

Proposed Method — Three-Phase Study

Phase 1

Baseline Evaluation

- Apply JPEG AI compression at BPP levels 0.12 – 1.00
- Evaluate CLIP-Det (3 score variants) on original vs. compressed images
- Measure AUC / Accuracy / F1 as a function of bitrate

Phase 2

Frequency-Domain Analysis

- 2D Fourier spectra of noise residuals
- 1D radial power profiles: real vs fake
- DCT coefficient distributions under compression
- Spectral distance vs AUC drop correlation

Phase 3

Mitigation Strategies

- Preprocessing (median, Gaussian, JPEG-75)
- Per-BPP logistic regression score fusion (calibration-based)
- Linear-probe fine-tuning of the Corvi2023 detection head

Dataset

OpenFake 20k

Balanced dataset of real and AI-generated face images, widely used as a deepfake detection benchmark.

Total images	~20,000
Classes	Real / Fake (1:1)
Resolution	Variable (RGB)
Source	Kaggle (AI-generated faces)
Test sample	200 images: 100 real + 100 fake

JPEG AI Compression Pipeline

- 1 Convert all images to PNG for codec compatibility
- 2 Compress with JPEG AI reference software (GitLab)
- 3 4 target BPP levels: 0.12 · 0.25 · 0.50 · 1.00
- 4 Reconstruct and verify all output files
- 5 Build comp_df manifest with measured BPP metadata

Target BPP levels:



Experimental Setup

Detector: CLIP-Det

ClipBased-SyntheticImageDetection (GRIP-UNINA)

CLIP features + linear classifier · 3 output scores:
clipdet_latent10k_plus, Corvi2023, fusion

Primary score: Corvi2023 (AUC > 0.75 on uncompressed data)

Framework & Environment

Language: Python

Frameworks: Pytorch, JPEG AI

Environment: Google Colab

Evaluation Metrics

Primary: **AUC** — threshold-independent, reliable

Secondary: Accuracy, F1-score (threshold: LLR > 0)
Reported per BPP level and per detector variant

Calibration / Test Split (Phase 3)

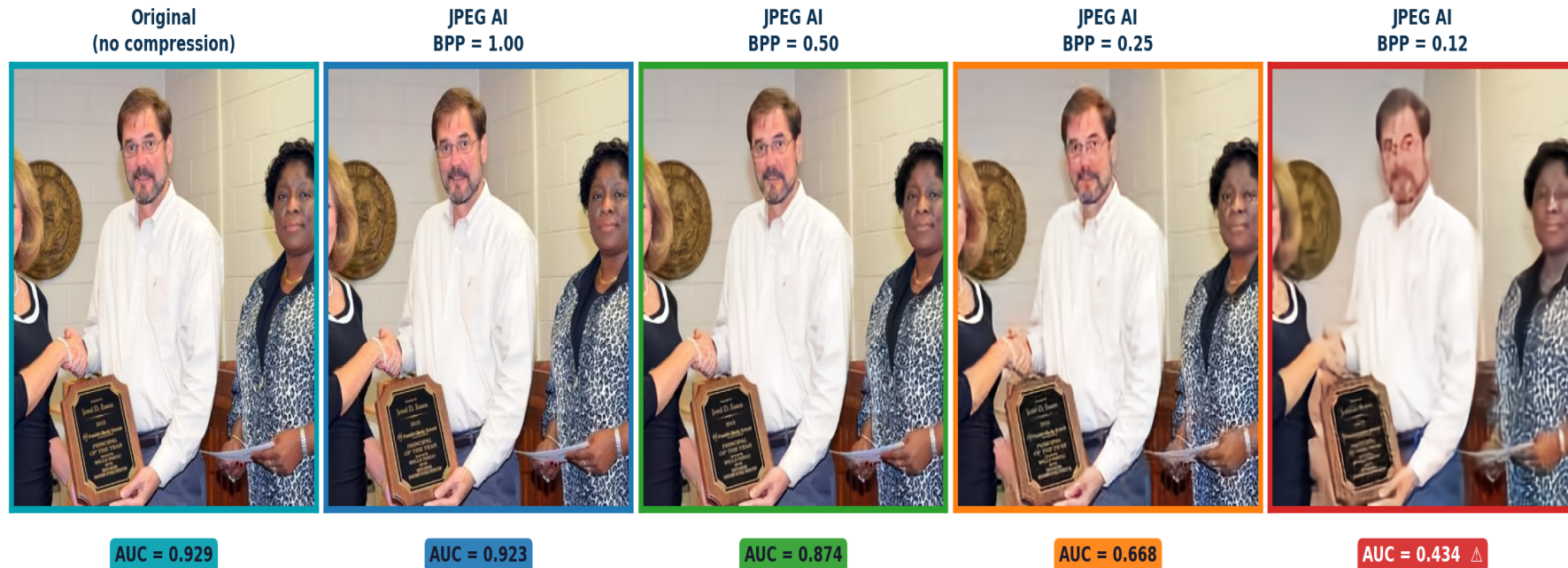
50% calibration / 50% test split (stratified, seed=42)

Calibration: used to fit Strategies B & C

Test: held-out evaluation — no data leakage

JPEG AI Reconstruction Quality

Real Image — JPEG AI Reconstruction Quality vs. Bitrate



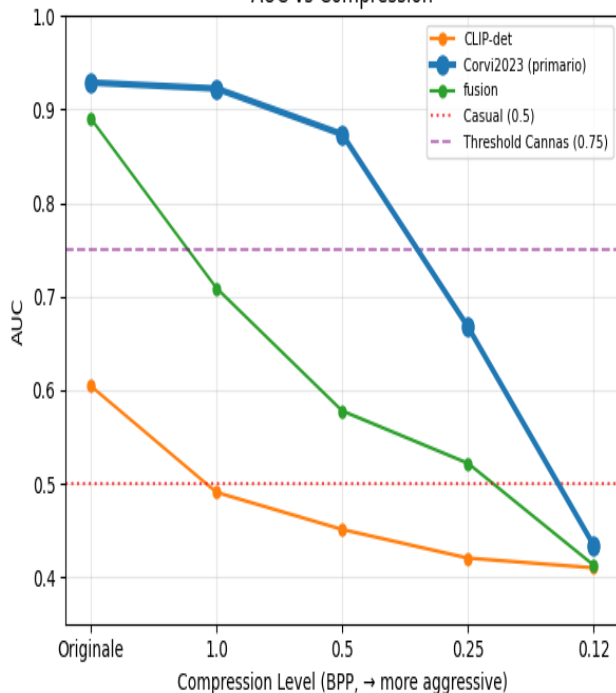
Real image (OpenFake dataset):

Artifacts become visible at BPP = 0.25 (blur, edge ringing). At BPP = 0.12 the neural codec fingerprint dominates — detector inverts (AUC = 0.434).

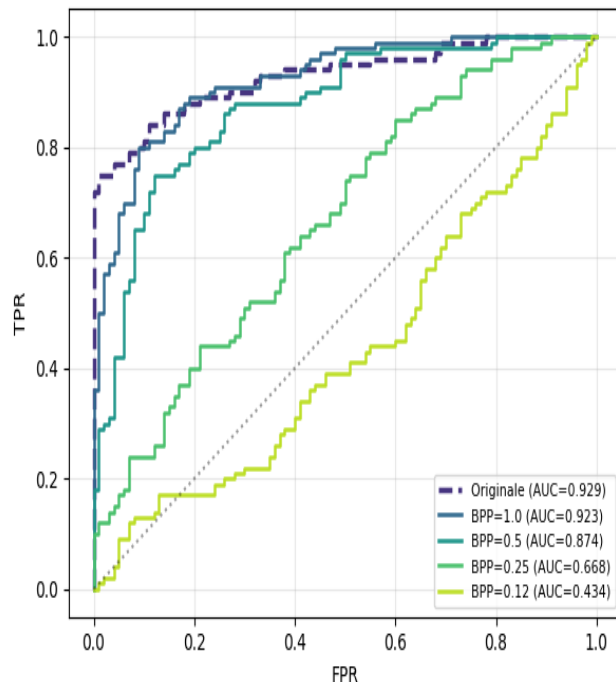
Phase 1 — Baseline Evaluation under Compression

Phase 1 — JPEG AI as threat to deepfake detection

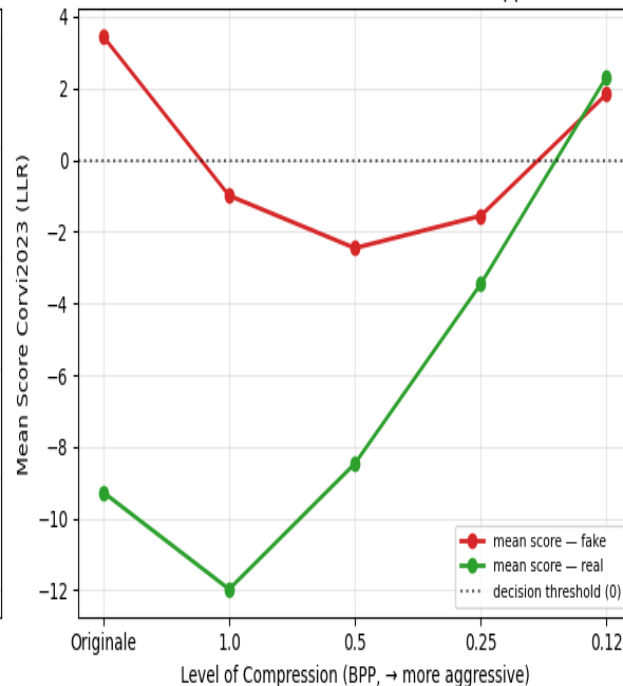
AUC vs Compression



Curves ROC — Corvi2023

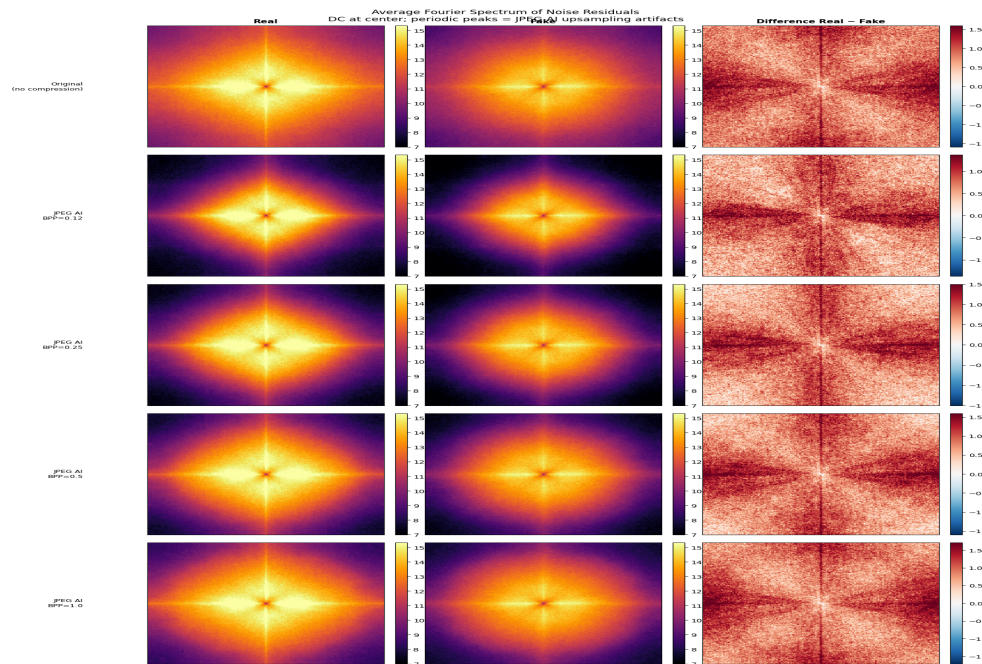


Inversion: at low BPP real/fake are swapped

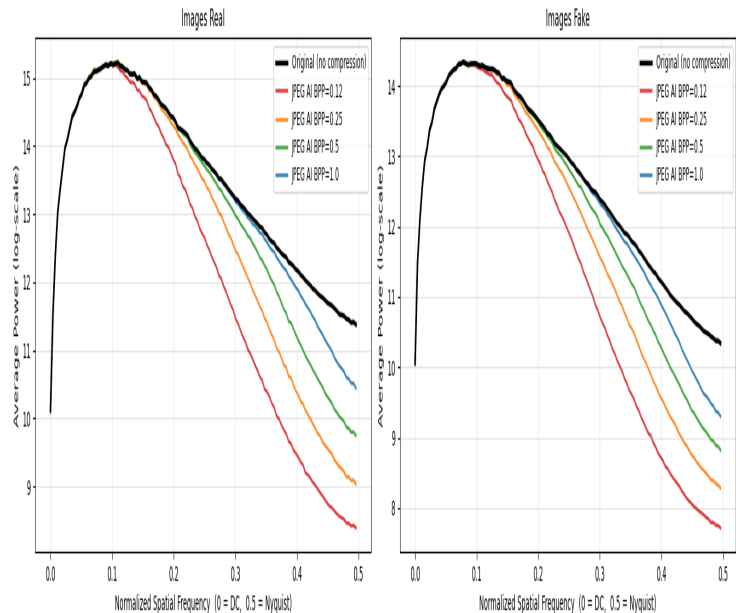


Key findings: AUC drops sharply at low BPP · Counter-forensic inversion at BPP=0.12 · Corvi2023 is the most resilient detector

Phase 2 — Frequency Analysis: Fourier Spectra



Radial Profile of the Power Spectrum of Noise Residuals
(peaks = periodicity of the JPEG AI decoder)



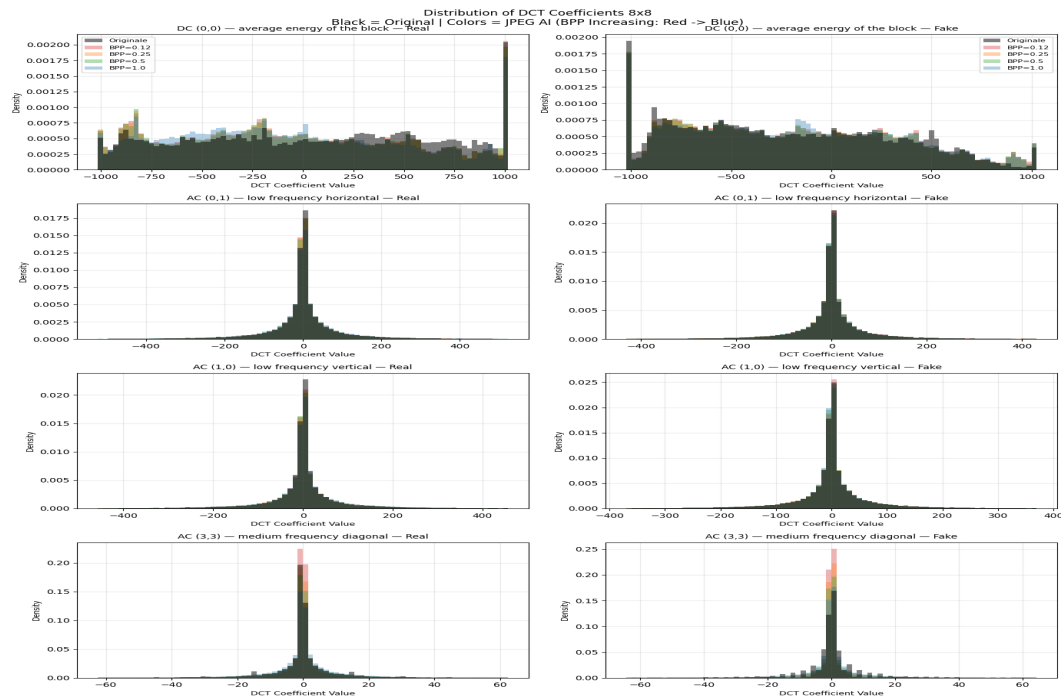
2D Average Power Spectrum

JPEG AI introduces periodic peaks from upsampling.
These artifacts are absent in original images.

Radial Power Profile

Real and fake spectral profiles converge at low BPP.
The detector discriminative signal collapses.

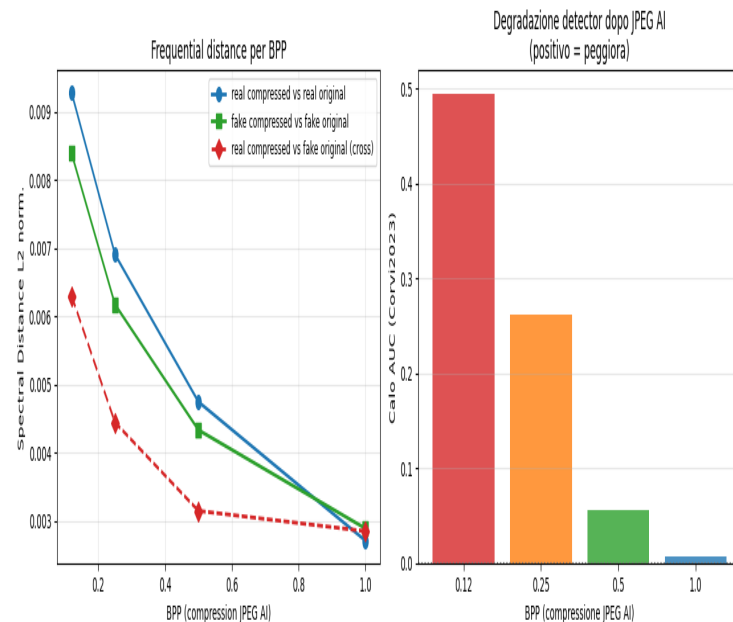
Phase 2 — DCT Coefficient Distributions & Spectral Distance vs. AUC Drop



8x8 DCT Coefficient Distributions

JPEG AI smooths high-frequency DCT coefficients differently from JPEG, erasing forensic cues detectors rely on.

Spectral Distance Introduced by JPEG AI and Detector Degradation



Spectral Distance vs. AUC Drop

Strong correlation: higher spectral perturbation predicts larger AUC degradation.

Phase 3 — Mitigation Strategies

Strategy A

Compression-Aware Preprocessing

Training-free · Applied before inference

median3: 3×3 median filter

gaussian: Gaussian blur ($\sigma = 0.8$)

jpeg75: JPEG re-encoding at quality 75

Aim: suppress JPEG AI neural fingerprints while preserving the detection signal.

Strategy B

Per-BPP Score Fusion (Logistic Regression)

Calibration-based · Low computational overhead

CLIP-Det outputs 3 complementary scores.

A logistic regression fitted per BPP level on calibration scores re-weights each detector.

Aim: restore discriminative power when one detector inverts under compression.

Strategy C

Linear-Probe Fine-Tuning

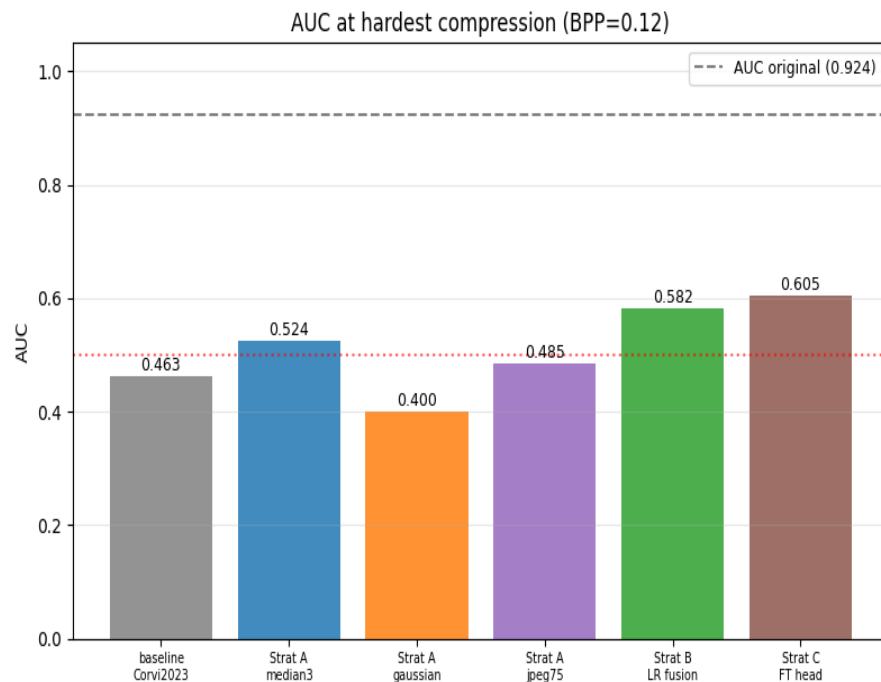
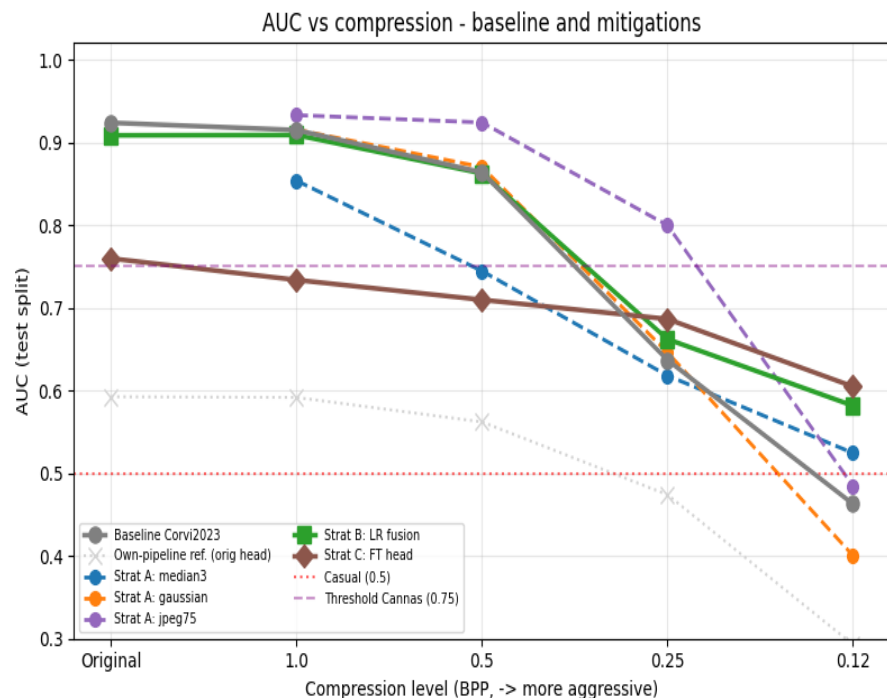
Calibration-based · Zero extra inference cost

ResNet50 backbone (Corvi2023) frozen. Only the Linear(2048→1) head retrained on multi-BPP augmented calibration data.

Aim: new decision boundary robust to JPEG AI feature distribution shift.

Phase 3 — Mitigation Results

Phase 3 - Mitigation strategies (primary detector: Corvi2023)



Main findings: Strategies B & C recover significant AUC at low BPP · Strategy A provides modest gains with zero training cost · Trade-off: effectiveness vs. computational overhead

Conclusions

1

JPEG AI significantly degrades deepfake detection

AUC drops from > 0.75 to below 0.5 at BPP = 0.12 — counter-forensic inversion.

2

Frequency analysis explains the mechanism

JPEG AI upsampling artifacts eliminate the spectral fingerprints that detectors rely on.

3

Mitigation is feasible

Strategies B & C recover 10–25% AUC at low BPP. Strategy A is competitive with zero training.

References

- [1] E. D. Cannas et al., "Is JPEG AI Going to Change Image Forensics?" ICCVW 2025, pp. 1575-1586.